

R SCRIPT FILE – PROJECT 2 GROUP 3 SECI 1143 PSDA SECTION 02

Hypothesis Testing Two-Sample

```
ridership_data <- read.csv("ridership_headline.csv", header=TRUE)
lrt_ampang <- na.omit(ridership_data$rail_lrt_ampang)
mrt_kajang <- na.omit(ridership_data$rail_mrt_kajang)
var.test(lrt_ampang, mrt_kajang)
t.test(lrt_ampang, mrt_kajang, alternative = "two.sided", var.equal = FALSE)
var.test(lrt_ampang, mrt_kajang)
View(ridership_data)
savehistory("~/SECPH UTM/SECI1143 Probability & Statistical Data Analysis/1. R
Materials/Public Transport Ridership/Hypothesis Testing Two-Sample.Rhistory")
```

Correlation Testing

```
correlation_analysis <- cor(ridership_headline$rail_mrt_kajang,
ridership_headline$rail_lrt_kj)
plot(ridership_headline$rail_mrt_kajang, ridership_headline$rail_lrt_kj)
```

Console For Regression

```
# ---- 0. Folders & data -----
> work_dir <- "C:/Users/hariz/OneDrive/Desktop/STATS/PSDA Project 2"
> img_dir <- file.path(work_dir, "Images")
> dir.create(img_dir, showWarnings = FALSE, recursive = TRUE) # make Images/ if missing
> setwd(work_dir)
>
> ride <- read.csv("ridership_headline.csv", stringsAsFactors = FALSE)
> ride$date <- as.Date(ride$date)
>
> # Keep only the rows where BOTH variables are recorded
> dat <- ride[!is.na(ride$rail_lrt_ampang) & !is.na(ride$rail_lrt_kj),
+           c("date", "rail_lrt_ampang", "rail_lrt_kj")]
>
> # Show the sample size and clean data
> cat("Sample size (n):", nrow(dat), "\n")
Sample size (n): 2708
> summary(dat[, c("rail_lrt_ampang", "rail_lrt_kj")])
rail_lrt_ampang rail_lrt_kj
Min. : 6587  Min. : 7195
1st Qu.: 95205 1st Qu.: 116982
Median : 144048 Median : 172135
Mean : 140994  Mean : 186375
3rd Qu.: 198979 3rd Qu.: 276786
Max. : 270317  Max. : 352504
> # ---- A1. Scatter plot with fitted line -----
> plot(dat$rail_lrt_ampang, dat$rail_lrt_kj,
+      pch = 20, col = rgb(0, 0, 1, 0.25),
+      main = "LRT Kelana Jaya vs LRT Ampang daily ridership",
+      xlab = "x : LRT Ampang ridership (passengers/day)",
+      ylab = "y : LRT Kelana Jaya ridership (passengers/day)")
> abline(model, col = "red", lwd = 2)
Error: object 'model' not found

> # ---- A2. Correlation coefficient (r) -----
> ct <- cor.test(dat$rail_lrt_ampang, dat$rail_lrt_kj)
> print(ct)           # Highlight r ~ 0.966, p < 2.2e-16 in the console
```

Pearson's product-moment correlation

```
data: dat$rail_lrt_ampang and dat$rail_lrt_kj
t = 194.67, df = 2706, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9634981 0.9685253
sample estimates:
cor
```

0.9661032

```
> # ---- A3. Fit the least-squares regression line y = b0 + b1*x -----
> model <- lm(rail_lrt_kj ~ rail_lrt_ampang, data = dat)
> summary(model)      # gives b0, b1, R-squared, std error, t-test on slope
```

Call:

```
lm(formula = rail_lrt_kj ~ rail_lrt_ampang, data = dat)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-140366 -11510   -914   11855   60727
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.207e+04  1.164e+03 -18.95  <2e-16 ***
rail_lrt_ampang 1.478e+00  7.594e-03  194.67  <2e-16 ***
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 23820 on 2706 degrees of freedom

Multiple R-squared: 0.9334, Adjusted R-squared: 0.9333

F-statistic: 3.79e+04 on 1 and 2706 DF, p-value: < 2.2e-16

```
>
```

```
> b0 <- coef(model)[1] # intercept ~ -22072
> b1 <- coef(model)[2] # slope    ~ 1.478
> cat(sprintf("Estimated line: y_hat = %.2f + %.4f * x\n", b0, b1))
Estimated line: y_hat = -22071.68 + 1.4784 * x
```

```
>
```

```
> # ---- A4. R^2 and standard error of estimate -----
> r2 <- summary(model)$r.squared      # ~ 0.933
> se <- summary(model)$sigma         # std error of estimate ~ 23817
> cat(sprintf("R-squared = %.4f (r = %.4f)\n", r2, sqrt(r2)))
R-squared = 0.9334 (r = 0.9661)
> cat(sprintf("Standard error of estimate (s) = %.1f\n", se))
Standard error of estimate (s) = 23817.4
```

```
>
```

```
> # ---- A5. t-test on the slope (H0: beta1 = 0 vs H1: beta1 != 0) -----
> coef(summary(model)) # Point out: Estimate / Std.Error / t value / Pr(>|t|)
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -22071.682007 1.164470e+03 -18.95427 2.503095e-75
rail_lrt_ampang  1.478402 7.594303e-03 194.67252 0.000000e+00
> confint(model)      # 95% CI for b0 and b1 (slope CI ~ [1.464, 1.493])
              2.5 %      97.5 %
(Intercept) -24355.023344 -19788.340669
rail_lrt_ampang  1.463511   1.493293
```

```
>
```

```
> # ---- A7. Residual diagnostics (assumptions) -----
> par(mfrow = c(1, 2))
> plot(model, which = 1) # Residuals vs Fitted: linearity & constant variance
> plot(model, which = 2) # Normal Q-Q: normality of the errors
> par(mfrow = c(1, 1))
>
```

Console for Hypothesis Testing Two-Sample

```
ridership_data <- read.csv("ridership_headline.csv", header=TRUE)
> lrt_ampang <- na.omit(ridership_data$rail_lrt_ampang)
> mrt_kajang <- na.omit(ridership_data$rail_mrt_kajang)
> var.test(lrt_ampang, mrt_kajang)

      F test to compare two variances

data:  lrt_ampang and mrt_kajang
F = 0.541, num df = 2676, denom df = 2676, p-value < 2.2e-16
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.5015135 0.5836001
sample estimates:
ratio of variances
    0.5410022

> t.test(lrt_ampang, mrt_kajang, alternative = "two.sided", var.equal =
FALSE)

Welch Two Sample t-test

data:  lrt_ampang and mrt_kajang
t = -11.585, df = 4915.9, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -26588.34 -18891.84
sample estimates:
mean of x mean of y
 140353.3  163093.4

> View(ridership_data)
```

Console For Analysis

```
> correlation_analysis <- cor(ridership_headline$rail_mrt_kajang,  
ridership_headline$rail_lrt_kj)  
> correlation_analysis  
[1] 0.9091434  
> plot(ridership_headline$rail_mrt_kajang,ridership_headline$rail_lrt_kj)
```